Exam 2 MTH 207 (Fall 2026) — J. Baggett

Score / 100

First Name	Last Name

Before you start. Write your name above and check that you have all 6 pages. All electronics off and out of sight — phones included, even just to tell the time, even in airplane mode. No calculators, notes, or books.

TO EARN POINTS YOU MUST SHOW YOUR WORK. Explain your reasons, justify your steps, and present your logic in an organized way. An answer with no supporting work earns no credit.

You have exactly 55 minutes. When you are told to stop you must stop **immediately** — that is in fairness to everyone. Keep your eyes on your own paper.

Geometry, if you need it. You are not expected to memorize these.

Circle	$A = \pi r^2,\ C = 2\pi r$	Sphere	$V = \frac{4}{3}\pi r^3,\ S = 4\pi r^2$
Square	$A = s^2$	Cylinder	$V = \pi r^2 h$
Rectangle	$A = \ell w$	Cone	$V = \frac{1}{3}\pi r^2 h$
Triangle	$A = \frac{1}{2}bh$	Cube	$V = s^3,\ S = 6s^2$
Right triangle	$a^2 + b^2 = c^2$		

1. [2 pts ea.] Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) Differentiating xy with respect to x gives $x\dfrac{dy}{dx} + y$.

(b) If f has an absolute minimum value at c, then $f'(c) = 0$.

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2. [7 pts] Evaluate the limit with **justifications**. Simplify until you have a number.

$$\lim_{x \to 0} \frac{\sin 5x}{3x^2 - 2x}$$

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3. [9 pts] A table of values for f, g, f' and g' is given. If $h(x) = f(g(x))$, find $h'(2)$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	4	2	5
2	1	6	3	7
3	7	8	1	9

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4. [8 pts] Use a linear approximation to estimate $\sqrt[3]{8.5}$. **State the function and the point you choose**, and show your work. Leave your answer as an exact fraction.

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5. [8 pts] If y depends on x according to $x^3 + y^3 = 6xy$, find $\dfrac{dy}{dx}$.

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6. Find the derivative of each function below.

(a) *(5 pts)* $f(x) = \dfrac{2x^2 - 1}{x^3 + 4}$

(b) *(5 pts)* $y = 4^x \log_3(x)$

(c) *(5 pts)* $y = x^{\cos x}$

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7. Derive the following derivative rules.

(a) *(7 pts)* Derive the derivative rule for $y = \sec x$. (You may use the derivative rules for sine and cosine.)

(b) *(7 pts)* Derive the derivative rule for $y = \tan^{-1} x$. (You may use the derivative rules for the trig functions, but **not** any derivative rules for inverse trig functions.)

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8. [15 pts] Water is poured into a cone-shaped tank at a rate of 2 cubic feet per minute. The cone points downward; it is 10 feet tall and its circular top has radius 5 feet. How fast is the depth of the water rising when the water is 4 feet deep? **Include units**, and leave your answer exact — a π in it is fine.

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9. [12 pts] Let $f(x) = 2x^3 - 3x^2 - 12x + 1$. Find the absolute maximum and absolute minimum values of f on the interval $[-2, 3]$, and say where each occurs.

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10. [8 pts] Verify that $f(x) = x^3 - x$ on $[0, 2]$ satisfies the hypotheses of the Mean Value Theorem (clearly state why each hypothesis is satisfied) and find all numbers c that satisfy the conclusion of the Mean Value Theorem. Give **exact** values.

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